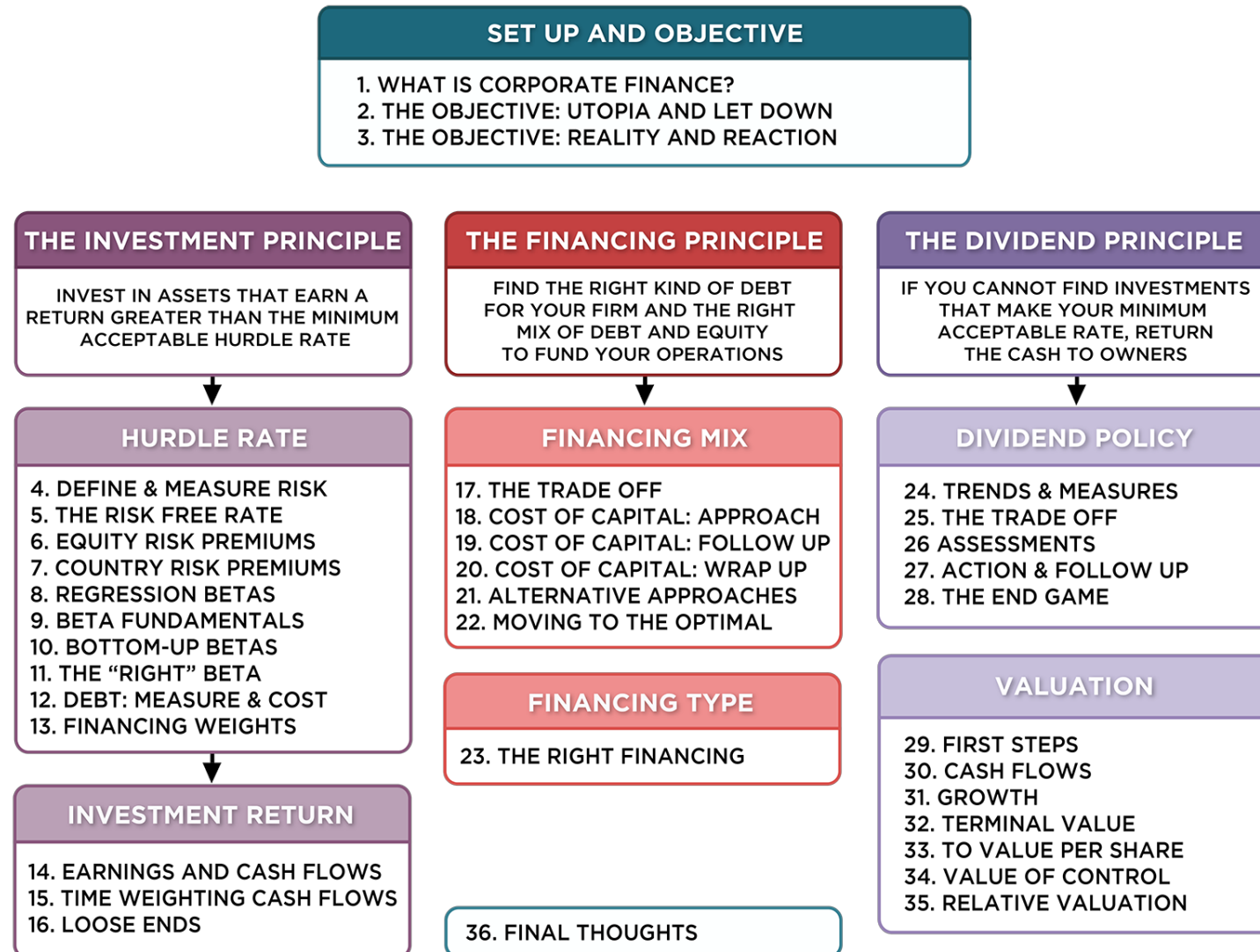


SESSION 08: REGRESSION BETAS

Stocks are risky! Really!

SESSION 08: REGRESSION BETAS



ESTIMATING BETA

- The standard procedure for estimating betas is to regress stock returns (R_j) against market returns (R_m):

$$R_j = a + b R_m$$

where a is the intercept and b is the slope of the regression.

- The slope of the regression corresponds to the beta of the stock, and measures the riskiness of the stock.
- The R squared (R^2) of the regression provides an estimate of the proportion of the risk (variance) of a firm that can be attributed to market risk. The balance ($1 - R^2$) can be attributed to firm specific risk.

ESTIMATING PERFORMANCE

- The intercept of the regression provides a simple measure of performance during the period of the regression, relative to the capital asset pricing model.

$$R_j = R_f + b (R_m - R_f)$$

$$R_j = R_f (1 - b) + b R_m$$

CAPITAL ASSET PRICING MODEL

$$R_j = a + b R_m$$

REGRESSION EQUATION

- If
 - $a > R_f (1 - b)$ STOCK DID BETTER THAN EXPECTED
 - $a = R_f (1 - b)$ STOCK DID AS WELL AS EXPECTED
 - $a < R_f (1 - b)$ STOCK DID WORSE THAN EXPECTED
- The difference between the intercept and $R_f (1-b)$ is Jensen's alpha. If it is positive, your stock did perform better than expected during the period of the regression.

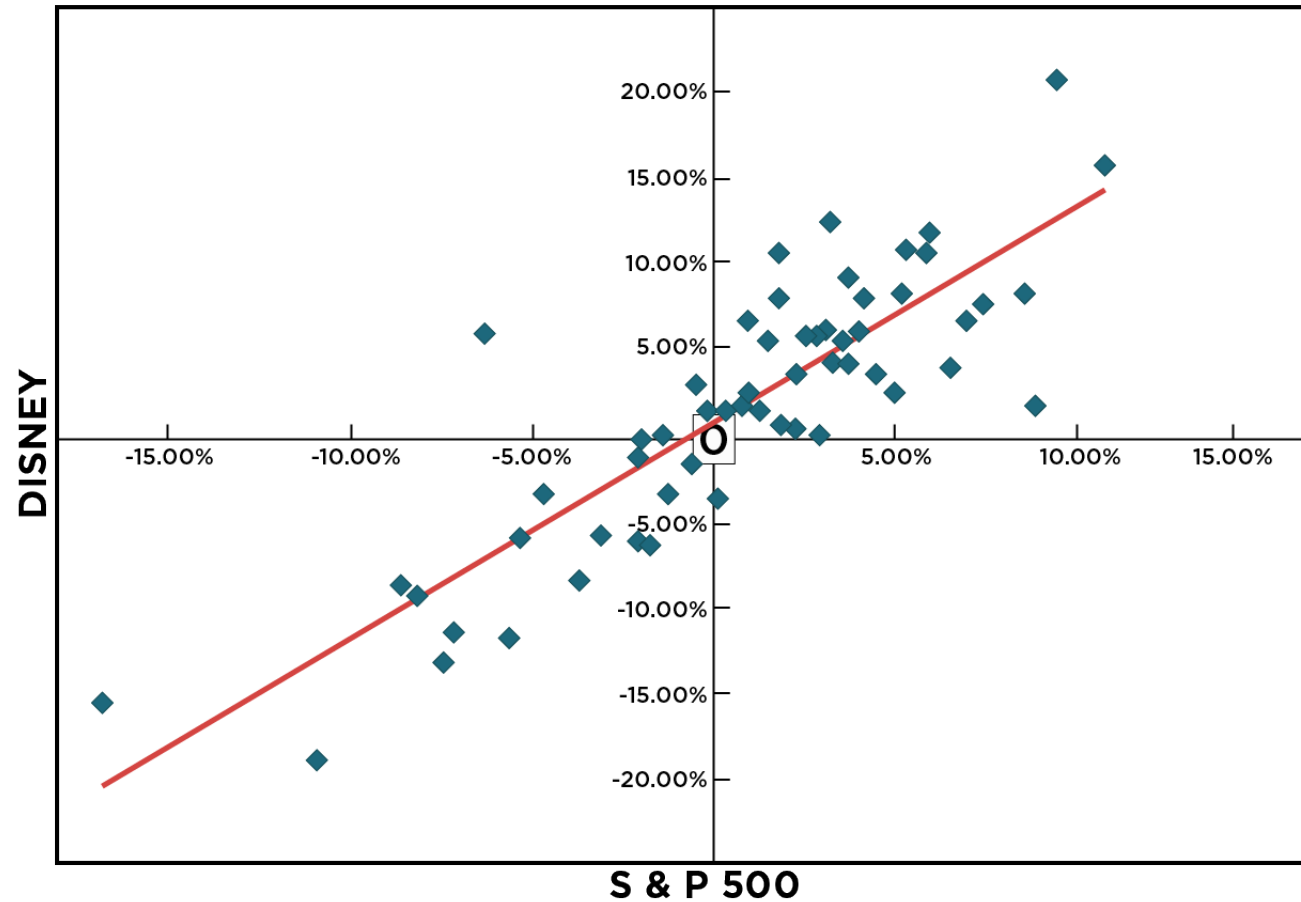
SETTING UP FOR THE ESTIMATION

- Decide on an estimation period
 - Services use periods ranging from 2 to 5 years for the regression
 - Longer estimation period provides more data, but firms change.
 - Shorter periods can be affected more easily by significant firm-specific event that occurred during the period.
- Decide on a return interval – daily, weekly, monthly
 - Shorter intervals yield more observations, but suffer from more noise.
 - Noise is created by stocks not trading and biases all betas towards one.
- Estimate returns (including dividends) on stock
 - $\text{Return} = (\text{Price}_{\text{End}} - \text{Price}_{\text{Beginning}} + \text{Dividends}_{\text{Period}}) / \text{Price}_{\text{Beginning}}$
 - Included dividends only in ex-dividend month
- Choose a market index, and estimate returns (inclusive of dividends) on the index for each interval for the period.

CHOOSING THE PARAMETERS: DISNEY

- Period used: 5 years
- Return Interval = Monthly
- Market Index: S&P 500 Index.
- For instance, to calculate returns on Disney in December 2009,
 - Price for Disney at end of November 2009 = \$ 30.22
 - Price for Disney at end of December 2009 = \$ 32.25
 - Dividends during month = \$0.35 (It was an ex-dividend month)
 - $\text{Return} = (\$32.25 - \$30.22 + \$0.35) / \$30.22 = 7.88\%$
- To estimate returns on the index in the same month
 - Index level at end of November 2009 = 1095.63
 - Index level at end of December 2009 = 1115.10
 - Dividends on index in December 2009 = 1.683
 - $\text{Return} = (1115.1 - 1095.63 + 1.683) / 1095.63 = 1.78\%$

DISNEY'S HISTORICAL BETA



$$\text{RETURN ON DISNEY} = .0071 + (1.2517)(\text{RETURN ON MARKET})$$

(0.10)

$$R^2 = 0.73386$$

ANALYZING DISNEY'S PERFORMANCE

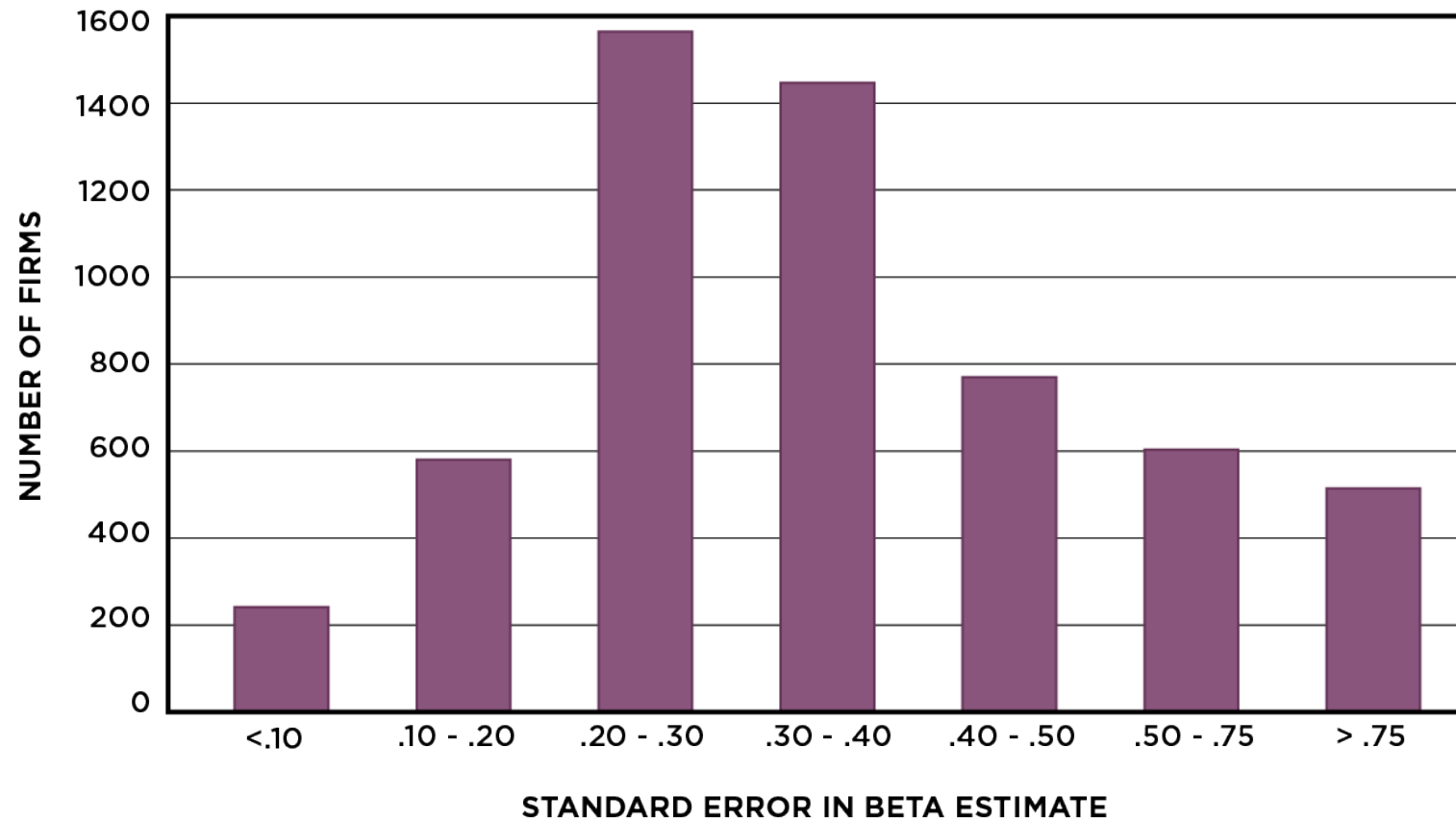
- Intercept = 0.712%
 - This is an intercept based on monthly returns. Thus, it has to be compared to a monthly riskfree rate.
 - Between 2008 and 2013
 - Average Annualized T.Bill rate = 0.50%
 - Monthly Riskfree Rate = $0.5\%/12 = 0.042\%$
 - Riskfree Rate (1-Beta) = $0.042\% (1-1.252) = -0.0105\%$
- The Comparison is then between
 - Intercept versus Riskfree Rate (1 - Beta)
 - 0.712% versus 0.0105%
 - Jensen's Alpha = $0.7122\% - (-0.0105)\% = 0.723\%$
- Disney did 0.723% better than expected, per month, between October 2008 and September 2013
 - Annualized, Disney's annual excess return = $(1.00723)^{12} - 1 = 9.02\%$
- This positive Jensen's alpha is a sign of good management at the firm.
 - True
 - False

ESTIMATING DISNEY'S BETA

- Slope of the Regression of 1.25 is the beta
- Regression parameters are always estimated with error. The error is captured in the standard error of the beta estimate, which in the case of Disney is 0.10.
- Assume that I asked you what Disney's true beta is, after this regression.
 - What is your best point estimate?
 - What range would you give me, with 67% confidence?
 - What range would you give me, with 95% confidence?

THE DIRTY SECRET OF “STANDARD ERROR”

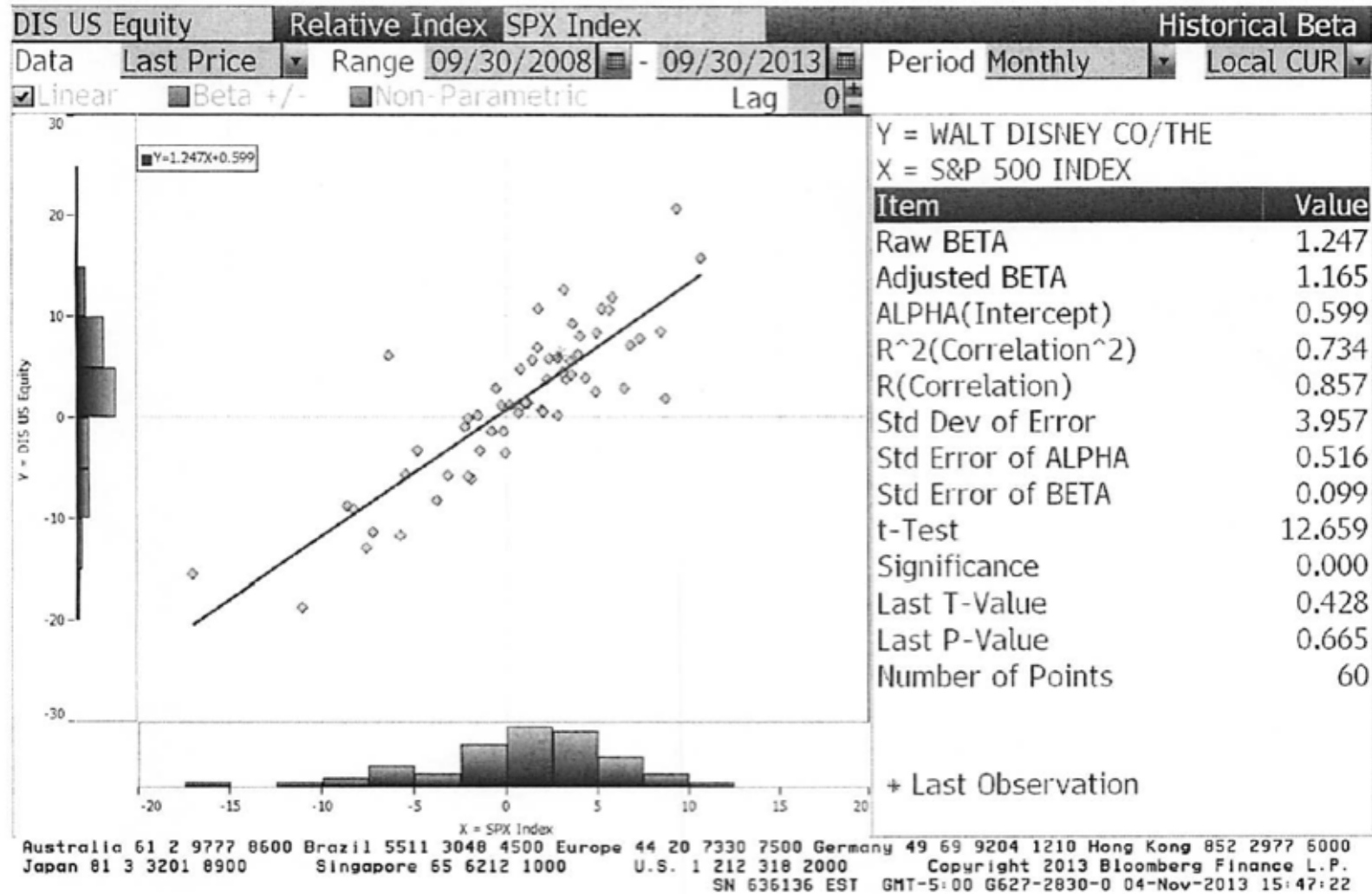
DISTRIBUTION OF STANDARD ERRORS: BETA ESTIMATES FOR U.S. STOCKS



BREAKING DOWN DISNEY'S RISK

- R Squared = 73%
- This implies that
 - 73% of the risk at Disney comes from market sources
 - 27%, therefore, comes from firm-specific sources
- The firm-specific risk is diversifiable and will not be rewarded.
- The R-squared for companies, globally, has increased significantly since 2008. Why might this be happening?
- What are the implications for investors?

BETA ESTIMATION: USING A SERVICE (BLOOMBERG)



ESTIMATED EXPECTED RETURNS FOR DISNEY IN NOVEMBER 2013

- Inputs to the expected return calculation
 - Disney's Beta = 1.25
 - Riskfree Rate = 2.75% (U.S. ten-year T-Bond rate in November 2013)
 - Risk Premium = 5.76% (Based on Disney's operating exposure)

$$\begin{aligned}\text{Expected Return} &= \text{Riskfree Rate} + \text{Beta} (\text{Risk Premium}) \\ &= 2.75\% + 1.25 (5.76\%) \\ &= 9.95\%\end{aligned}$$

USE TO A POTENTIAL INVESTOR IN DISNEY

- As a potential investor in Disney, what does this expected return of 9.95% tell you?
 - This is the return that I can expect to make in the long term on Disney, if the stock is correctly priced and the CAPM is the right model for risk,
 - This is the return that I need to make on Disney in the long term to break even on my investment in the stock
 - Both
- Assume now that you are an active investor and that your research suggests that an investment in Disney will yield 12.5% a year for the next 5 years. Based upon the expected return of 9.95%, you would
 - Buy the stock
 - Sell the stock

HOW MANAGERS USE THIS EXPECTED RETURN

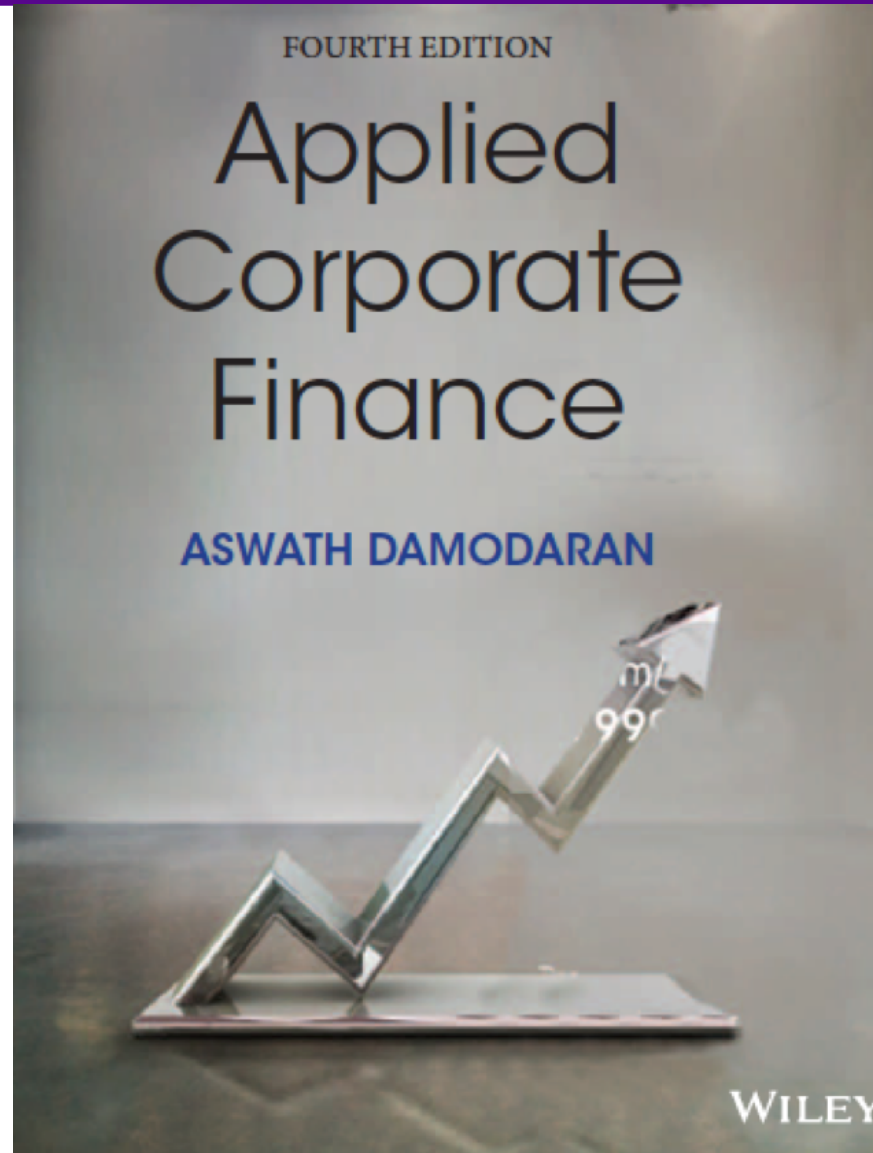
- Managers at Disney
 - need to make at least 9.95% as a return for their equity investors to break even.
 - this is the hurdle rate for projects, when the investment is analyzed from an equity standpoint
- In other words, Disney's cost of equity is 9.95%.
- What is the cost of not delivering this cost of equity?



APPLICATION TEST: ANALYZING THE RISK REGRESSION

- If you can get a beta regression page (or output) for your company against a market index, answer the following questions:
 - How well or badly did your stock do, relative to the market, during the period of the regression?
 - $\text{Intercept} - (\text{Riskfree Rate}/n) (1 - \text{Beta}) = \text{Jensen's Alpha}$
where n is the number of return periods in a year (12 if monthly; 52 if weekly)
 - What proportion of the risk in your stock is attributable to the market? What proportion is firm-specific?
 - What is the historical estimate of beta for your stock? What is the range on this estimate with 67% probability? With 95% probability?
 - Based upon this beta, what is your estimate of the required return on this stock? $\text{Riskless Rate} + \text{Beta} * \text{Risk Premium}$

Task
Break down
the beta
regression for
your company



Optional: Read
Chapter 4